Supporting Ultra-High-Resolution Digital Agriculture Tasks with Fully Synthetic Curriculum Learning

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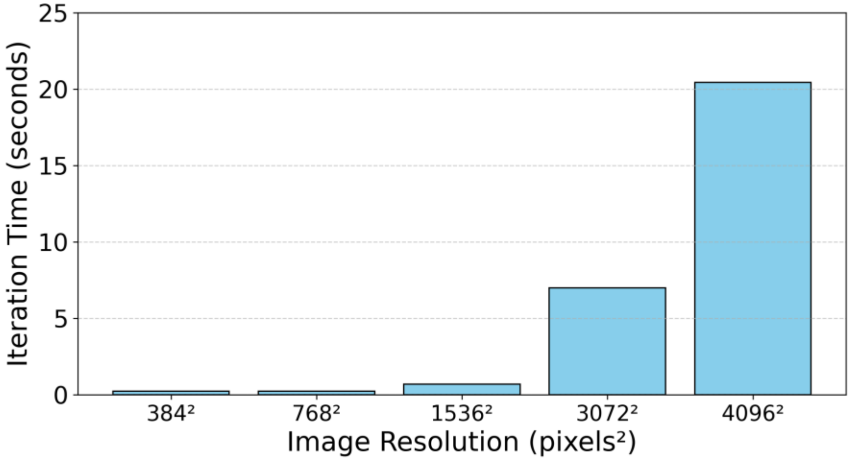
Introduction

- Vision Transformers (ViT)
  - Vision Transformers are increasingly popular in computer vision
    - Commonly studied within NSF ICICLE Institute for yield estimation, disease phenotyping, pest control, among other tasks.
  - ViT models are very expensive to train from scratch
  - Therefore, computer vision researchers ***pretrain*** ViT models on large, general, **low-resolution** datasets
    - Expensive upfront cost
    - Inexpensive, domain specific fine-tuning.
- We seek to enable high-quality high-resolution ViT modeling at a lower training cost.

Motivation

- Most pretrained transformers have a very low resolution
  - Generally, 224^2 or 384^2
  - High-resolution pretraining is very expensive!
  - Even "high-resolution" models do not have true 4K resolution support
    - Generally, $1{,}000^2$
- Agronomists frequently collect 4K and higher resolution images for use in digital agriculture
- Low-resolution pretrained models are unable to fully utilize the rich structure of these ultra-high-resolution images.
  - The accuracy in remote sensing, digital pathology, medical imaging, and others can depend on fine-grained features or long-ranged dependencies
  - These features can be suppressed when down-sampling or cropping high-resolution images.



What you see What the model sees

Motivation

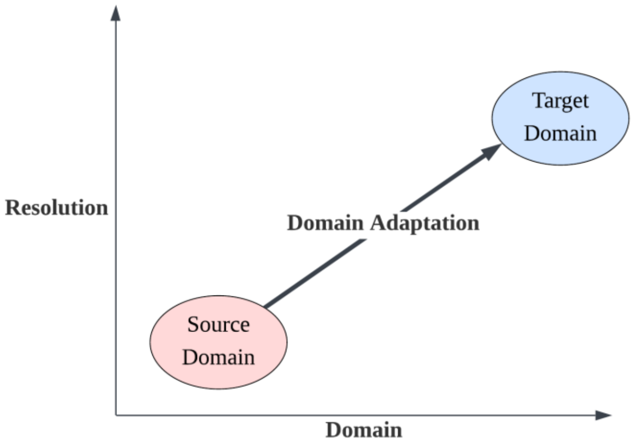
Down-sampling

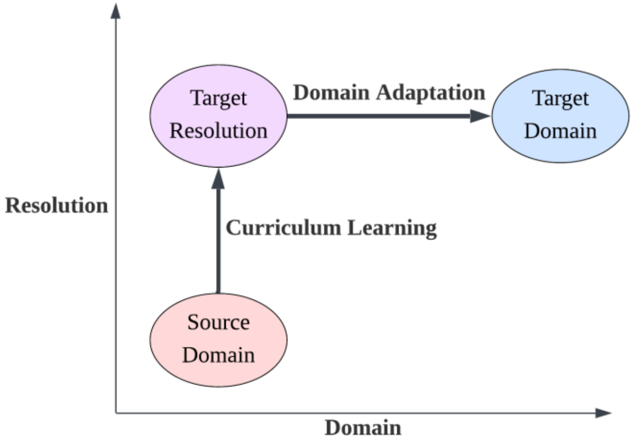
What you see

What the model sees

Motivation

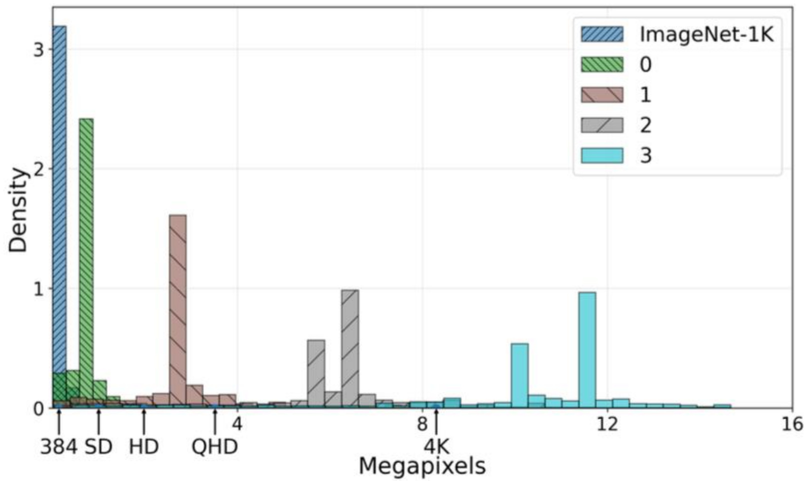
- Typically, Low-resolution pretrained models are finetuned directly at high-resolution, which requires the model to learn both:
  - A domain shift from generic web images to agriculture.
  - A shift from low to ultra-high resolution.
- This complexity forces us to train on more high-resolution samples
  - Using more samples adds cost to training and risks overfitting
  - Using too few samples risks underfitting the model
- Therefore, digital agriculture and other fields need an efficient, reusable, and general method to extend pretrained models to ultra-high resolution.





Method

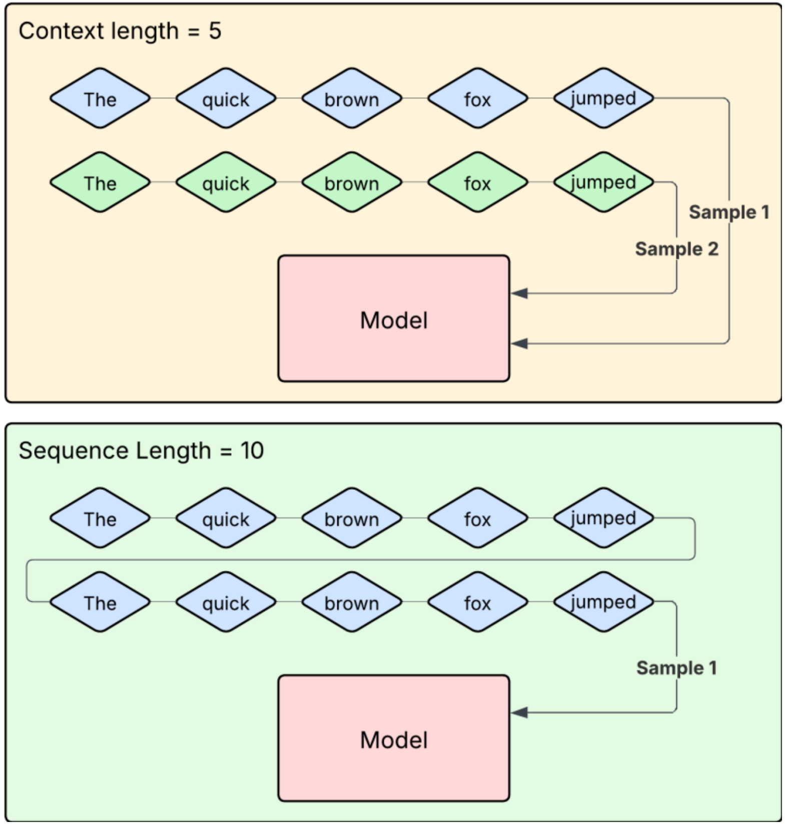
- Curriculum Learning
  - We borrow Curriculum Learning from natural language processing (NLP)
  - Curriculum learning attempts to mimic human learning, starting with easy problems and progressively increasing the *difficulty*.
    - This slow introduction allows smoother training
  - In our case this *difficulty* is the image resolution, also called the context window
  - We will progressively increase the resolution of the images throughout training.
- We use a staged curriculum learning approach, aiming to smoothly warm the model to higher resolutions.
  - 384^2 (pretraining resolution), 768^2, 1536^2, 3072^2, 4096^2 (target resolution)



Method

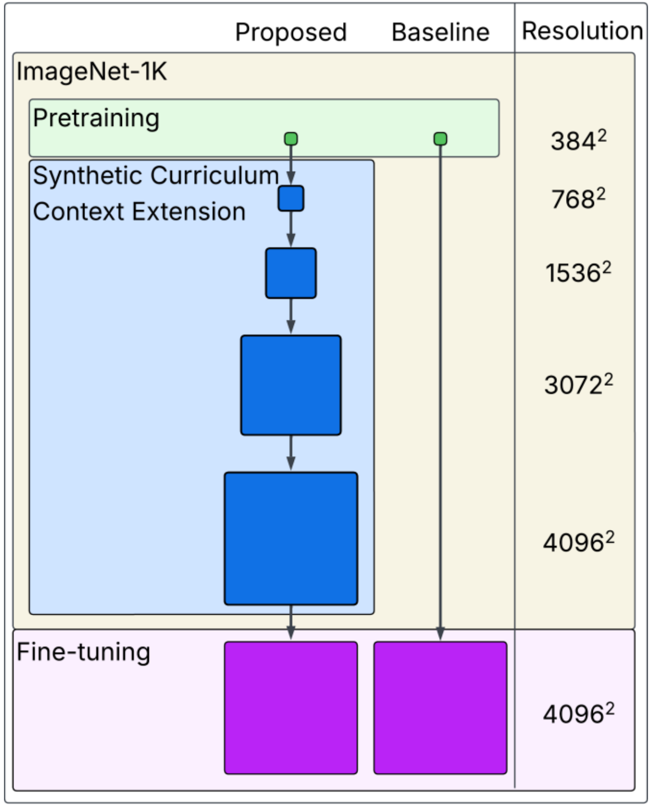
- Curriculum learning in NLP
  - NLP can employ curriculum learning by concatenating tokens to create longer sequences
  - No change in distribution - Just more tokens!
- However, context extension in computer vision is not so easy!
- Pretraining datasets such as ImageNet-1k contain mostly low-resolution images
  - We cannot just use more tokens
  - We need larger images!
- Naively resizing images degrades data quality

In text, context extension is easy!



Method

- Synthetic data preparation
  - We use a super-resolution model: EDSR
  - Use EDSR to create a staged synthetic dataset for continued pretraining
    - Matches the data distribution of the original pretraining dataset.
  - EDSR allows us to increase the size of images while matching the pretraining distribution and preserving data quality.
- Use synthetic data and curriculum learning to finetune pretrained models for high-resolution tasks
- The resulting model can be treated as a pretrained model and finetuned for any domain-specific high-resolution computer vision task.
- We validate our method by fine-tuning and evaluating our model on downstream digital agriculture tasks.



Evaluation

- We evaluate our model on two downstream ultra-high-resolution digital agriculture datasets.
  - Images were taken with phone cameras in fields at a resolution of 4K.
- We use two datasets:
  - the Corn Disease dataset: 11,402 images
  - Soybean Disease dataset: 6,126 images
- These datasets were created to develop models to identify and classify crop diseases, deficiencies, or other stressors.

Class	Train	Val
Healthy	2233	558
Common Rust	252	63
Corn Borer	366	91
Grey Leaf Spot	721	180
Herbicide Sensitivity	1034	258
Holcus Spot	46	11
Mg or K Deficiency	72	18
N Burn	1798	449
N Deficiency	855	214
NCLB	765	191
P Deficiency	392	98
Potassium Deficiency	590	147
Total	9124	2278

Class	Train	Val	Total
BBP	214	53	267
Dicamba Damage	623	156	779
FELS	957	239	1196
Healthy	1284	321	1605
Insect Damage	956	269	1225
SDS	800	254	1054
Total	4834	1292	6126

Evaluation

- We finetune three different models:
  - ViT-384, a low-resolution model trained by resizing the image
  - ViT-4096, a high-resolution model trained directly from the low resolution pretrained model
  - **ViT-Synth (ours)**, the model trained using our proposed high-resolution synthetic curriculum learning strategy.

Evaluation

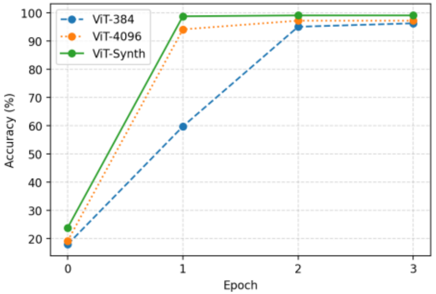
- Our evaluation demonstrates consistent improvements in accuracy, precision, recall, and F1 score when adapting models for high-resolution digital agriculture tasks.
- ViT-4096 underperformed compared to the low-resolution baseline
  - Without further adaptation, training high-resolution models directly from low-resolution pretrained models results in performance degradation.

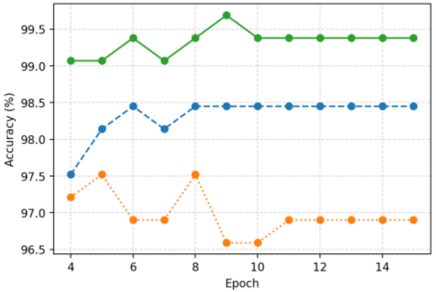
Model	**Acc.**	**Prec.**	**Recall**	**F1**
ViT-384	98.95	0.989	0.958	0.971
ViT-4096	96.90	0.977	0.976	0.976
ViT-Synth (Ours)	**99.38**	**0.995**	**0.996**	**0.995**

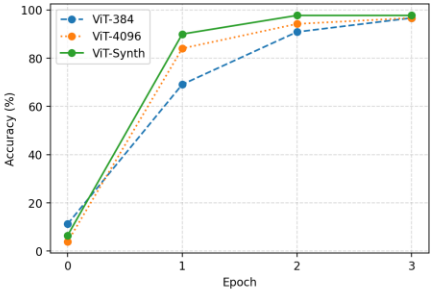
Model	**Acc.**	**Prec.**	**Recall**	**F1**
ViT-384	98.45	0.987	0.978	0.982
ViT-4096	98.25	0.959	0.959	0.959
ViT-Synth (Ours)	**99.65**	**0.998**	**0.997**	**0.997**

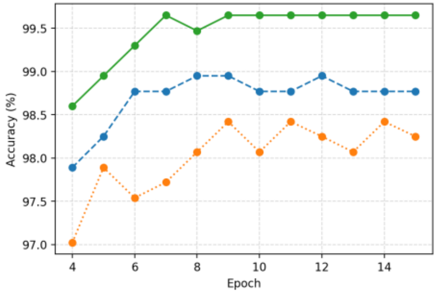
Evaluation

- Models trained with our synthetic curriculum learning also have faster time to convergence and improved stability throughout finetuning for both the Soybeans (left) and Corn (right) datasets
- Therefore, we produce models that are more adaptable and generalizable to high-resolution tasks, decreasing the costs of producing high-resolution models









Discussion

Limitations

- Created synthetic images will degrade in quality when using pretraining datasets that are out of distribution for EDSR
  - Our method is not specific to EDSR!
  - We can mitigate degradation by choosing a super-resolution model that performs well on the targeted pretraining dataset.
- Super-resolution models will also degrade in performance as the target resolution increases, introducing more artifacts.
  - We mitigate this effect by using **stages** of synthetic data
  - Creating one stage of high-resolution data that is resized to smaller images would mean lower resolution images may contain artifacts only found in more extreme super-resolution
  - Using super-resolution in stages minimizes artifacts introduced at higher resolutions by EDSR
  - However, super-resolution models will not be able to support arbitrarily high resolutions without degrading performance

Conclusion

- We improve model performance at high-resolution by up to 2.48% and 1.2% when compared to the high-resolution and low-resolution baselines, respectively.
- Our synthetic curriculum learning strategy extends the applicability of low-resolution pretrained models for high-resolution computer vision tasks.
- Our method is orthogonal to model architecture and pretraining dataset choices.
- In future work, we aim to strengthen the contributions of this method by analyzing the number of stages used, the target resolution limit, and other important factors for context length extension in vision transformers.

Thank You!

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